Don Bosco Institute of Technology, Mumbai

(An Autonomous Institute Affiliated to the University of Mumbai)

Mumbai - 400070

Department of Computer Engineering

[PROJECT TITLE]

A Report Submitted

in partial fulfillment of the requirements for

COMMUNITY ENGAGEMENT PROJECT 1

Course Code: 25CE3CEP01

Semester III, Academic Year 2026-27

Submitted by

Name of Student	Roll No.	Student ID / UID
[Enter details]	[Enter details]	[Enter details]
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[Enter details]	[Enter details]	[Enter details]

Supervised by

[Name of Faculty Mentor]

AY 2026-27

CERTIFICATE

This is to certify that the Community Engagement Project 1 entitled **"[PROJECT TITLE]"** is a bona fide work carried out by the following students of Second Year Computer Engineering, Semester III, during Academic Year 2026-27, in partial fulfillment of the requirements of the course Community Engagement Project 1 (25CE3CEP01).

Student Name	Roll No.	Signature

_____	_____	_____
Signature	Signature	Signature
Faculty Guide	**Head of Department**	**Principal**

COMMUNITY ENGAGEMENT PROJECT 1 REPORT APPROVAL

This report titled "[PROJECT TITLE]" submitted by

1. ..
2. ..
3. ..
4. ..

is approved for completion of Community Engagement Project 1 (25CE3CEP01), Semester III, Academic Year 2026-27, Department of Computer Engineering.

Examiners

1. _____

2. _____

Date:

Place: Kurla, Mumbai

DECLARATION

We hereby declare that the project report titled "[PROJECT TITLE]" is our original work carried out under the guidance of [Faculty Mentor]. The work has not been submitted elsewhere for the award of any degree or academic credit. All sources of information, data, images, code, standards and published material used in this report have been appropriately acknowledged.

Student Name	Roll No.	Signature

VISION, MISSION AND PROGRAMME SPECIFIC OUTCOMES

Institute Vision

Shaping a sustainable tomorrow - DBIT will be known to have an innovative, enjoyable and holistic learning environment that transforms individuals into socially conscious citizens the Don Bosco way and will lead in research and entrepreneurship in the area of sustainable technologies.

Institute Mission

Engineering Excellence with Purpose

1. To create future engineers who work with honesty and integrity and excel in the use of technology for the benefit of the underprivileged.
2. To train engineers to be innovative problem solvers and entrepreneurs who engage in research and life-long learning.
3. To provide a diverse and stimulating environment for staff and students to grow holistically

Department Vision

To become a center of academic excellence in the field of computer science and engineering for developing technically competent professionals in the service of society.

Department Mission

1. Develop an academic system towards effective teaching learning process by ensuring enthusiastic participation from the stakeholders for consistent improvement through evaluation and feedback.
2. Promote undergraduate collaborative research in the emerging areas of computer science.
3. Provide learning environment for the students to enhance problem solving skills with professional and ethical responsibilities.
4. Prepare industry ready professionals with social commitment.

Programme Specific Outcomes (PSOs)

PSO 1: The Computer Engineering graduate will be able to demonstrate the ability to understand, analyze and develop computer applications in the field of Computer Science and Engineering.

PSO 2: The Computer Engineering graduate will be able to demonstrate leadership, professional communication, ethical practices, teamwork and social commitment.

PSO 3: The Computer Engineering graduate will be able to demonstrate research skills and utilize modern computing tools to solve real life problems.

ABSTRACT

Instruction: *Summarize the community need addressed, the proposed solution and methodology, the key results and validation, and the impact on the identified community/stakeholders. (300 words max)*

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CHAPTER 1

INTRODUCTION

1.1 Problem Identification and selection

Instruction: *Establish the socio-technical background of the chosen community domain and why technological intervention is required in this sector. Document the real-world issue observed on the ground, highlighting who the affected stakeholders are and how their everyday life is hindered.*

1.2 Empathy Study & Stakeholder Analysis

Instruction: *Present the empathy map diagrams across the Says, Thinks, Does, and Feels quadrants. Detail the constructed user personas, explicitly capturing their demographics, goals, daily frustrations, and functional needs.*

1.3 Motivation

Instruction: *Articulate the ethical, social, and academic drive for tackling this community challenge through engineering methodologies.*

1.4 Problem Statement

Instruction: *Frame the formal, concise 1–2 sentence problem statement describing the target user, the exact pain point, and the proposed digital solution.*

1.5 Objectives and Scope

Instruction: *Enumerate 3–5 measurable technical objectives and clearly define what features fall within this phase versus what is out of scope.*

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Literature Survey on Current Trends

Instruction: *Summarize 5 recent research papers in a comparative table outlining author details, methodologies, key findings, and technical limitations.*

Ref. No.	Paper Name and Year	Author(s)	Summary of Paper	Research Gap Identified
[1]				
[2]				
[3]				
[4]				
[5]				

Table No. 2.1: Literature Survey on Current Trends

2.2 Comparative Analysis of Existing Products / Systems

Instruction: *Evaluate 5 current mobile applications or web platforms attempting to address this issue, charting their operational pros and cons.*

Ref. No.	Name of App/Website	Features	Missing Features	Comparison with proposed solution
[6]				
[7]				
[8]				
[9]				
[10]				

Table No. 2.2: Feature comparison of existing and proposed products / systems

2.3 Research Gaps and Identified Limitations

Instruction: *Critically highlight where current academic research and market solutions fail to adequately satisfy community requirements.*

2.4 Feature List and Justification

Instruction: ***T****abulate the preliminary feature set proposed for your system and provide a written engineering justification based on the identified gaps.*

Name of app/website:		
Sr. No.	**Features**	**Justification**
1.		
2.		

CHAPTER 3

FIELD RESEARCH AND SURVEY METHODOLOGY

3.1 Methodology and Data Collection Plan

Instruction: *Describe the sampling strategy, geographic context, and structured data-collection procedure used to engage with the target community.*

3.2 Questionnaire and Interview Design

Instruction: *Present the faculty-approved questionnaire (minimum 20 questions across sections) and justify how each question informs system design.*

3.3 Survey and Interview Findings Analysis

Instruction: Document the reach of the survey (targeting 150–200 responses) or video interviews, summarizing respondent demographics like age, role, and location. Analyze user responses using charts and percentages, verifying whether the community validates the problem and accepts the suggested features.

3.4 Community Feature Acceptance

Instruction: Present the revised, evidence-backed list of system features refined directly from user feedback and interview suggestions.

CHAPTER 4

SYSTEM DESIGN AND CREATIVE IDEATION

4.1 Proposed System Architecture & Solution Approach

Instruction: *Provide a high-level architectural diagram illustrating user touchpoints, front-end workflows, and system interactions.*

4.2. UML Modelling

Instruction: *Present standard UML Use Case, Activity, and State Machine diagrams depicting actors, system processes, and operational state transitions with descriptive explanations.*

4.2.1 Use Case Diagram and Actor Descriptions

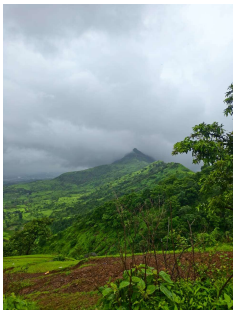


Figure 2.1: Sample Image

In Figure 2.1, it shown that…..

4.2.2 Activity Diagram and Process Flow

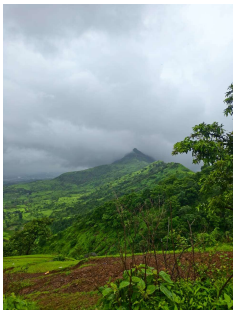


Figure 2.2: Sample Image 2

In Figure 2.2, it shown that…..

4.2.3 State Diagram and Transitions

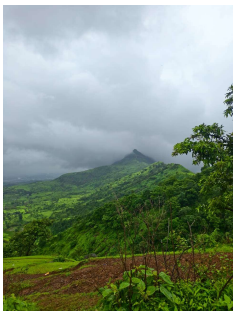


Figure 2.3: Sample Image 3

In Figure 2.3, it shown that…..

4.3 Project Timeline & Work Breakdown

Instruction: *Include a structured Gantt chart detailing project phase, work distribution among team members, and completion milestones.*

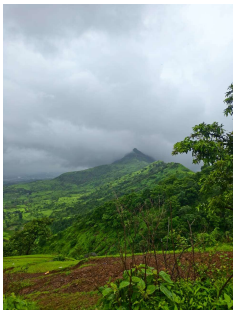


Figure 2.4: Sample Image 4

In Figure 2.4, it shown that…..

CHAPTER 5

PROTOTYPE IMPLEMENTATION AND PROOF OF CONCEPT

5.1 Development Environment & Software Tools Selected

Instruction: *State the chosen implementation tool (Figma, Canva, or HTML/CSS/JS) and justify its selection for building the proof of concept.*

5.2 Screen-by-Screen Walkthrough & Visual Design

Instruction: *Map the chronological 5-screen user journey (Landing, Selection, Action Form, Review, and Success Screen) that guides the community member. Document the primary trigger buttons across all screens, explaining their destinations and forward interaction paths without dead ends. Include high-resolution screenshots for each interface state alongside a brief explanation of screen purpose, visual elements, and user actions.*

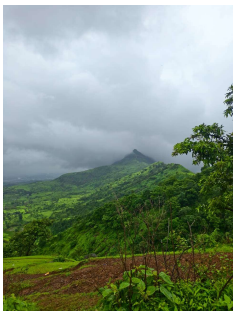


Figure 5.1: Sample Image 5

In Figure 5.1, it shown that…..

CHAPTER 6

USER TESTING, PEER REVIEW, AND DESIGN EVALUATION

6.1 Testing Methodology and Task Scenarios

Instruction: *Describe the task scenarios assigned to peer evaluators and the testing protocol followed during the walkthrough session.*

6.2 Usability Walkthrough Evaluation

Instruction: *Report the quantitative findings from the 35-point peer review rubric assessing clarity, button affordance, navigation flow, contrast, and consistency.*

6.3 Peer Constructive Feedback Analysis

Instruction: *Summarize the qualitative observations collected from peers, categorizing specific likes, criticisms, and points of confusion.*

CHAPTER 7

CONCLUSION: SOCIETAL IMPACT AND FUTURE SCOPE

7.1 Community Impact and Societal Value Created

Instruction: *Explain the practical, tangible benefits and long-term positive changes this engineered solution offers to community stakeholders. Outline the functional boundaries, technical constraints, or edge cases not handled in this prototype phase.*

7.2 Future Enhancements and Deployment Roadmap

Instruction: *Provide a realistic roadmap outlining backend integration, database setup, mobile deployment, and scalability plans. Provide a concluding synthesis summarizing project achievements from empathy-driven discovery to a validated, working prototype.*

REFERENCES

Instruction: *Use IEEE reference style. Cite all papers, books, standards, websites, datasets, software libraries, images and other sources used. Number references in the order cited.*

[1] Paper Reference in IEEE format

APPENDICES

Appendix A - Stakeholder Interaction / Community Evidence

Instruction: *Attach or summarize relevant* ***screenshot*** *evidence such as survey/interview questionnaire,* ***anonymized*** *responses, meeting/visit record, stakeholder feedback, permission/consent where applicable, and photographs with captions. As required by the course guidelines, include suitable documentary evidence of interaction, consultation or feedback from at least one external stakeholder (e.g., a letter, e-mail or acknowledgement from the stakeholder/organisation, or a dated meeting/visit record).*

Appendix B - Course Outcomes, CO-PO-PSO Mapping and Evidence of CO Achievement

Course Outcomes	Project Evidence / Artefact	Report Section
CO1: Identify community needs, stakeholder requirements, Sustainable Development Goals (SDGs), and relevant Knowledge and Attitude Profiles (KAPs) associated with the selected project. (Remember – Factual Knowledge)	• Completed 4-Quadrant Empathy Map (Says, Thinks, Does, Feels) • Detailed User Persona profile (Demographics, Needs, Frustrations, Goals)	Sections 1.1 and 1.2
CO2: Explain the social, environmental, technological, economic, and ethical factors influencing the development and implementation of community-oriented engineering solutions. (Understand – Conceptual Knowledge)	• Comparative literature survey matrix of 5 recent research papers • Benchmark analysis of 5 existing apps/systems highlighting limitations and research gaps	Sections 2.1 and 2.2
CO3: Apply appropriate engineering tools, techniques, methodologies, and technologies to develop and improve solutions addressing identified community needs. (Apply – Procedural Knowledge)	• UML system models (Use Case Diagram, Activity Diagram, and State Machine Diagram). • Work breakdown structure and Project Timeline Gantt Chart	Sections 4.2 and 4.3
CO4: Analyse system performance, stakeholder feedback, implementation challenges, and operational constraints to determine the effectiveness of the proposed solution. (Analyse – Conceptual and Procedural Knowledge)	• Structured field survey data analysis report (charts and percentage distributions from 150–200 respondents). • Formulated 1–2 sentence validated Problem Statement and justified feature list	Section 3.4 and Appendix A

Evaluate the functionality, sustainability, reliability, inclusiveness, and societal impact of the developed solution using appropriate technical and community-benefit criteria. (Evaluate – Metacognitive Knowledge)	• Completed 35-point Usability & Visual Walkthrough rubric across 7 heuristic parameters. • Structured Peer Constructive Feedback report (Likes, Criticisms, Friction points, Improvement ideas)	Sections 6.2 and 6.3
CO6: Design and validate an innovative, sustainable, and socially relevant engineering solution that addresses community needs while demonstrating responsible professional attitudes and relevant KAP attainment. (Create – Metacognitive Knowledge)	• Interactive 5-screen click-through Proof of Concept (PoC) prototype screenshots and forward CTA flows. • Public interactive prototype screenshots of demonstration (Figma) or version-controlled GitHub repository URL	Section 5.2

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2		1			1	2			1	1	1
CO2	2	2	2	1	1			1	2			1	1	1
CO3	2	2	3	2	2			1	2	1	1	2	2	2
CO4	2	3	3	2	2			1	2	1	1	2	2	2
CO5	2	2	3	2	3	1	1	1	2	2	1	2	2	2
CO6	2	2	3	2	3	1	1	1	2	2	2	3	2	2

(1 = Low, 2 = Medium, 3 = High, Blank = No Mapping)

CO-PO-PSO Mapping Justification

CO	Mapped PO(s) / PSO(s)	Justification for Mapping
CO1	PO1, PO2, PO3, PO8, PO9, PSO1, PSO2, PSO3	Applies foundational engineering knowledge, analytical thinking, problem identification, teamwork and communication through empathy tools and community interaction.
CO2	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PSO1, PSO2, PSO3	Requires analysis of user research and translation of findings into a clear, actionable problem statement.
CO3	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Uses creative thinking, collaboration and appropriate tools to generate and explore solution concepts.
CO4	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Involves analysing user feedback, comparing alternatives and investigating design feasibility and impact.
CO5	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Requires evaluation and justification of the selected solution using usability, societal need and sustainability criteria.
CO6	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Involves development and demonstration of a conceptual prototype using appropriate software/tools and communication of technical and societal impact.

REPORT FORMATTING GUIDELINES

Instruction: *This page is for guidance only. Delete this page before submitting the report.*

Item	Requirement / Recommendation
Paper size	A4
Font	Times New Roman
Body text	12 pt; justified; 1.15 to 1.5 line spacing
Chapter title	16 pt bold, uppercase or title case; consistent throughout
Section heading	14 pt bold
Subsection heading	12 pt bold
Margins	Left 2.5 cm; Right 2.0 cm; Top 2.0 cm; Bottom 2.0 cm
Figures / Tables	Number chapter-wise or sequentially; provide meaningful captions and cite them in text
References	IEEE style recommended; all references cited in text
Page numbering	Roman numerals for preliminary pages (optional); Arabic numerals from Chapter 1
Similarity / originality	Students must submit original work and acknowledge all sources/tools used
Report length	Use the length required to document the project adequately; avoid unnecessary repetition or full datasheet reproduction